

C.I	Frequency
0-10	2
10-20	3
20-30	9
30-40	5
40-50	1

Formula:

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times C$$

$$\text{Median} = L + \frac{\frac{n}{2} - F_c}{f_c} \times C$$

$$\text{A.M} = \frac{\sum_{i=1}^n f_i x_i}{n}$$

Solve these following questions.

1. Compute A.M.
2. Compute Mode and Median.
3. Draw Ogive Curve and locate Median.
4. Compute Q_3, D_7, P_{89} .

Solution:

1. The formula for Arithmetic Mean (A.M) is:

$$\text{A.M} = \frac{\sum_{i=1}^n f_i x_i}{n}$$

Where:

- $x_i = \text{Midpoint of each class} = \frac{\text{Lower Bound} + \text{Upper Bound}}{2}$
- $f_i = \text{Frequency}$

C.I	f_i	Midpoint x_i	$f_i x_i$
0-10	2	5	10
10-20	3	15	45
20-30	9	25	225
30-40	5	35	175
40-50	1	45	45

$$\sum f_i = n = 2 + 3 + 9 + 5 + 1 = 20$$

$$\sum f_i x_i = 10 + 45 + 225 + 175 + 45$$

$$\text{A.M} = \frac{500}{20} = 25$$

2. The formula of Mode is:

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times C$$

Where:

- Modal Class = Class with highest frequency = 20–30
- L = 20 (lower boundary of modal class)

- $C = 10$ (class width)
- $f_m = 9$ (frequency of modal class)
- $f_{m-1} = 3$ (preceding frequency)
- $f_{m+1} = 5$ (succeeding frequency)
- $\Delta_1 = f_m - f_{m-1} = 9 - 3 = 6$
- $\Delta_2 = f_m - f_{m+1} = 9 - 5 = 4$

$$Mode = 20 + \frac{6}{6+4} \times 10 = 26$$

Mode = 26

The formula of Median is:

$$\text{Median} = L + \frac{\frac{n}{2} - F_c}{f_c} \times C$$

Where:

- Median Class = Class containing $\frac{n}{2} = \frac{20}{2} = 10$
- Cumulative Frequency:

C.I	Frequency	C.F
0-10	2	2
10-20	3	5
20-30	9	14
30-40	5	19
40-50	1	20

The cumulative frequency just greater than 10 is 14, so the Median Class is 20-30.

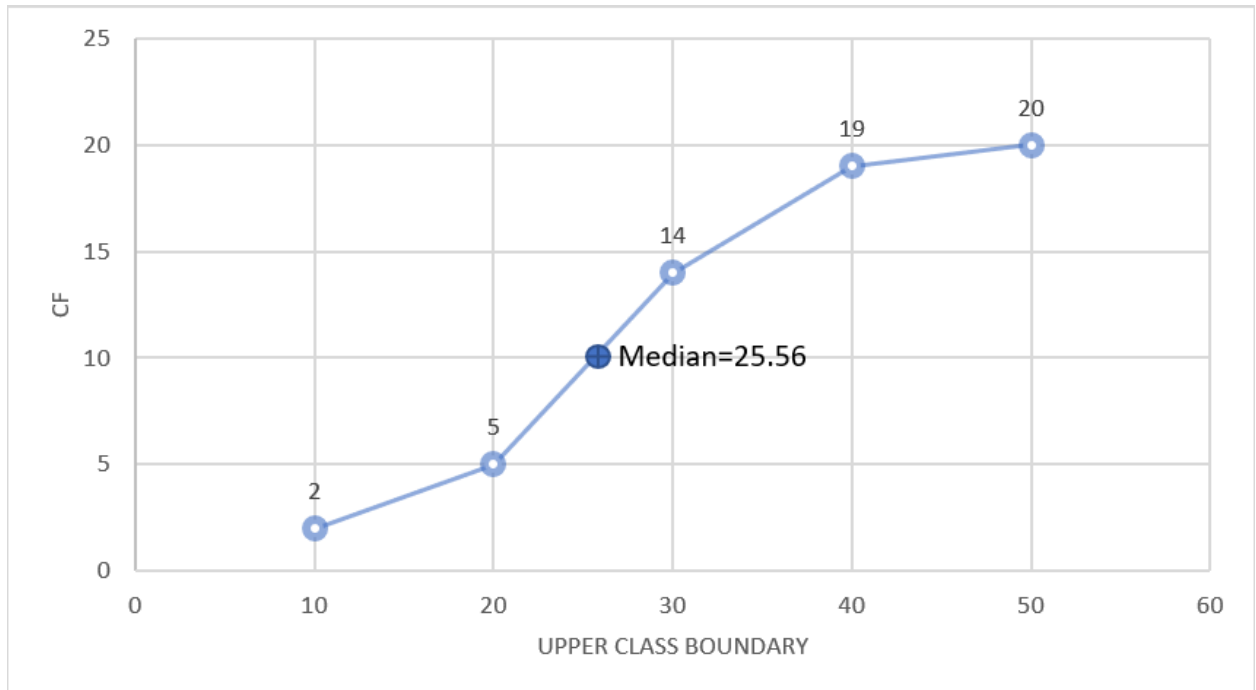
- $L = 20$ (lower boundary of median class)
- $C = 10$ (class width)
- $n = 20$
- $F_c = 5$ (cumulative frequency before median class)
- $f_c = 9$ (frequency of median class)

$$Median = 20 + \frac{10 - 5}{9} \times 10 = 25.56$$

Median = 25.56

3. Cumulative Frequency Table for Ogive Curve:

C.I	Frequency	CF	Upper Class Boundary
0-10	2	2	10
10-20	3	5	20
20-30	9	14	30
30-40	5	19	40
40-50	1	20	50



4. Q_3 Calculation (Third Quartile):

- Position: $\frac{3n}{4} = \frac{3 \times 20}{4} = 15$
- The cumulative frequency just greater than 15 is 19, so the Q_3 class is 30-40.
- $L = 30, F_c = 14, f_c = 5, C = 10$

$$Q_3 = 30 + \frac{15 - 14}{5} \times 10 = 32$$

D_7 Calculation (Seventh Decile):

- Position: $\frac{7n}{10} = \frac{7 \times 20}{10} = 14$
- The cumulative frequency just greater than 14 is 19, so the D_7 class is 30-40.
- Using the same class as Q_3 :

$$D_7 = 30 + \frac{14 - 14}{5} \times 10 = 30$$

P_{89} Calculation (89th Percentile):

- Position: $\frac{89n}{100} = \frac{89 \times 20}{100} = 17.8$
- The cumulative frequency just greater than 17.8 is 19, so the P_{89} class is 30-40.
- $L = 30, F_c = 14, f_c = 5, C = 10$

$$P_{89} = 30 + \frac{17.8 - 14}{5} \times 10 = 37.6$$